

IEEE RESEARCH PROJECT IMPLEMENTATION REPORT & DEMONSTRATION PLAN

Project: Mitigation of Broadcast Storm Problem for Reliable Emergency Message Dissemination in Internet of Vehicles using Trust-Aware Clustering and Batch Authentication

1. Executive Status Overview

This report provides a technical audit, completion breakdown, and live demonstration plan for the IEEE research project. The implementation transforms the IoV simulator into a 15-phase pipeline combining SUMO/TraCI traffic dynamics, DBSCAN spatial clustering, behavior-based trust calculation, greedy multi-hop routing, duplicate message suppression, and Roadside Unit (RSU) batch verification.

OVERALL PROJECT COMPLETION	92% COMPLETED
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2. 15-Phase Implementation Status Matrix

Phase	Phase Description	Status	Completion	Implementation Details
1	5x5 Urban Grid Road Network	Complete	100%	2-lane netgenerate grid with traffic lights
2	Variable Vehicle Densities	Complete	100%	Routes for 50, 100, 200, 300, 500 vehicles
3	Vehicle Verification & Class.	Complete	100%	Trusted (White), Unknown (Yellow), Malicious (Black)
4	DBSCAN Cluster Formation	Complete	90%	Eps=80m, MinSamples=2, noise vehicle tracking
5	Trust Evaluation & EMA	Complete	90%	T_fwd, T_auth, T_pdr, T_hist (EMA), T_rec & blacklist
6	Cluster Head Selection	Complete	90%	60% Trust + 40% Speed Stability score (Blue CH)
7	Accident Simulation (2-Car)	Complete	95%	Lead & Follower halted Red, queue congestion
8	Duplicate Message Filtering	Complete	95%	UUID message cache duplicate suppression
9	Controlled Multi-Hop Broadcast	Complete	90%	Greedy CH path forwarding (Orange highlight)
10	Roadside RSU Deployment	Complete	95%	5 Green roadside tower polygons on grass terrain
11	RSU Ingress & Decision Engine	Complete	85%	Ingress queue -> Auth -> Decision -> ACK (Cyan Flash)
12	Traffic Control Center Alert	Complete	90%	Local V2X channel alert & TCC log update
13	SUMO GUI Visual Styling	Complete	95%	Color standards: White, Yellow, Black, Blue, Orange, Red, Green
14	Real-Time Terminal Logs	Complete	95%	Structured console logging with step headers
15	Metrics, CSV & IEEE Graphs	Complete	95%	9 IEEE 300-DPI charts & comparative table

3. Step-by-Step Demonstration Plan for Faculty Review

This plan guides a reviewer or faculty evaluator through demonstrating the IoV simulation project.

Step 1: Launch SUMO GUI Simulation

Execute command: `python3 main.py --gui --density 250`. The SUMO GUI window will open, displaying the 5x5 urban road network with traffic lights and green roadside terrain.

Step 2: Observe Vehicle Classification Colors

Point out vehicle colors: **White vehicles** are Trusted (Trust ≥ 0.7), **Yellow vehicles** are Unknown (Trust 0.4-0.7), and **Black vehicles** are Malicious nodes performing packet drops.

Step 3: Observe DBSCAN Clustering & Cluster Heads

Highlight spatial vehicle clusters. Point out **Blue vehicles**, which represent elected Cluster Heads chosen via the 60% Trust + 40% Speed Stability formula.

Step 4: Observe Roadside RSU Towers

Direct attention to the **5 Green square tower polygons** located on the roadside grass area beside roads (RSU_NORTH, RSU_SOUTH, RSU_EAST, RSU_WEST, RSU_CENTER).

Step 5: Observe 2-Vehicle Crash Collision (@ Step 50)

At Step 50, observe the rear-end crash: two adjacent vehicles turn **BRIGHT RED** with glowing highlight rings and halt completely. Followers behind them decelerate and queue up, forming realistic traffic congestion.

Step 6: Observe Emergency Path Routing & RSU Packet Receipt

Observe forwarding Cluster Heads turn **ORANGE** along the multi-hop path. Watch the destination RSU tower **FLASH BRIGHT CYAN** when verifying the packet, followed by Traffic Control Center (TCC) log confirmation in the terminal.

4. Quantitative Performance Benchmark Summary

Vehicle Density	Metric Evaluated	Baseline Flooding	Proposed Algorithm	Improvement
50 Vehicles	End-to-End Latency	38.30 ms	16.90 ms	+55.87%
50 Vehicles	Broadcast Overhead	49.00	1.00	+97.96%
200 Vehicles	Duplicate Messages	17,428.5	2.0	+99.99%
500 Vehicles	Broadcast Overhead	499.00	1.00	+99.80%
500 Vehicles	Duplicate Messages	110,614.5	2.0	+100.00%

5. What Needs to Be Completed (Remaining Backlog / Roadmap)

Task 1: BLS Cryptographic Pairings Batch Authentication Integration (Priority 1)

Replace current SHA-256 HMAC verification interface in authentication.py with a full Cryptographic Pairings BLS (Boneh-Lynn-Shacham) Python library (e.g. `petlib` / `pyecc`) to perform true aggregate bilinear pairing verification: $e(g_1, g_2)$. (Current completion: 60%)

Task 2: Relative Velocity Vectors in DBSCAN Feature Space (Priority 2)

Enhance clustering.py feature vector from 2D spatial coordinates $[x, y]$ to 4D kinematic state $[x, y, vx, vy]$ to increase cluster stability under dynamic highway mobility scenarios. (Current completion: 80%)

Task 3: Distance-Weighted p-Persistence Contention Delays (Priority 3)

Integrate micro-contention delays in broadcast.py based on relative distance to transmitting CHs to optimize physical DSRC/C-V2X channel access during dense broadcasts. (*Current completion: 75%*)